

Aditya Sreenath

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SUMMARY

Data Engineer & ML Practitioner with 5+ years of production automation experience, currently pursuing an MS in Data Science at the University of Michigan. Focused on integrating predictive modeling and computer vision into production-ready cloud architectures.

EDUCATION

University of Michigan - Dearborn | GPA: 3.78/4.0

Dearborn, MI, USA

Master of Science in Data Science

Sep 2024 – Apr 2026

- **Relevant Graduate Coursework:** Probability and Statistical Model, Applied Regression Analysis, Deep Learning, Introduction to Big Data, Big Data Analysis and Visualization, Database Systems, Cloud Computing, Project Management

SRM Institute of Science and Technology | CGPA: 7.60/10.00

Chennai, TN, India

Bachelor of Technology in Software Engineering

June 2015 – May 2019

PROFESSIONAL EXPERIENCE

Automation & Data Engineer | Capgemini (Client: Synchrony Financial)

Dec 2019 - Aug 2024

- Engineered a stateful automation system processing **1,000+** daily marketing campaign emails across multi-stage pipelines, utilizing cross-referencing logic to maintain synchronization across disparate datasets.
- Developed ETL workflows to extract and validate data from **on-demand sources ranging from MBs to 5GB+** of diverse semi-structured files (PDF, PPT, XLSX, DOC).
- Architected an end-to-end validation system cross-referencing JIRA release data against QA documentation and standards for **up to 50 concurrent software releases** during peak periods ensuring high data integrity and production standards.
- Led the migration of legacy automation bots to cloud-ready A360 architecture, ensuring **zero downtime** for critical business processes.
- Standardized team-wide **engineering best practices** and provided technical mentorship for junior engineers during large-scale automation migrations.

SKILLS

Data Engineering: Apache Spark, Hadoop, Database Design & Normalization, Differential Privacy, ETL, Anonymization.

Machine Learning and AI: Generative AI (RAG, LLMs), YOLO (v5/v8), SARIMAX, LSTM, Random Forest, Logistic Regression, Pytorch.

Cloud (AWS): Kinesis Video Streams, Rekognition, Lambda, DynamoDB, API Gateway, S3, SNS, SQS.

Languages & Tools: Python (Pandas, PySpark, Numpy, Matplotlib, Scikit-learn), SQL, R, Git, JIRA, Automation Anywhere (A360), Microsoft Office, Power Bi.

Statistical Methods: Hypothesis Testing (T-tests, ANOVA, Chi-squared), Predictive Modeling, Regression Analysis, Differential Privacy.

PROJECT EXPERIENCE

AI Research Assistant

- Developed a Retrieval-Augmented Generation pipeline using **LangChain** and **Llama 3** to extract and synthesize information from unstructured research papers.
- Integrated **ChromaDB** to manage document embeddings, focusing on similarity search to improve the relevance and grounding of AI-generated answers.

Healthcare Data Privacy & Governance

- Engineered a **Differential Privacy** framework using **Laplace and Gaussian mechanisms** to secure 750k+ patient records against re-identification attacks while maintaining 99%+ analytical accuracy.
- Utilized **Root Mean Square Error (RMSE)** to quantify the privacy-utility trade-off, ensuring that noise levels (0.016) effectively masked sensitive attributes like admission deposits and BMI.

Smart Door Authentication System

- Architected a real-time facial recognition system using **AWS Kinesis Video Streams** and **Rekognition**, integrating **Lambda** functions and **API Gateway** to automate OTP-based access and owner authorization workflows.

Detroit Crime Analysis & Predictive Policing

- Processed **750k+ records** using **Pandas**, **Geopandas**, and **DBSCAN** clustering to engineer temporal and spatial features, identify crime hotspots, and develop interactive dashboards for temporal pattern analysis.
- Conducted a comparative analysis of SARIMAX and LSTM architectures to optimize forecasting accuracy ($R^2 = 0.47$).

CERTIFICATIONS

- AWS Certified Cloud Practitioner - BSSQX77K4BB1QC30
- Automation Anywhere Masters Certification - 14661583